1.Marico Case:

a. Explore what can be potential objectives in this case. Order them by importance.

- Improving service levels as its critical to retain customer trust and prevent revenue loss
- Optimize Warehouse Count seeing to that the service constrains are being met.
- Reduce Freight and Operational Costs as it has a direct impact on the profit without affecting (largely) the service levels.

b. What should Arun go with, 22 warehouses or 24 warehouses? Why?

Marico Ltd. serves four regions in India. North, East, West, and South, with varying service level improvements when increasing the number of warehouses from 22 to 24. The service levels improve significantly in some regions: North (75% to 85%, +10 percentage points), West (93% No change), and South (83% to 87%, +4 percentage points), while the East region shows a dip of 2% (65% to 63%). Assuming demand is proportional to population and economic activity, revenue contributions by region are estimated as ₹21 billion for North and West (30% each), and ₹14 billion for East and South (20% each). Using a simplified assumption that a 1 percentage point increase in service levels leads to a 0.5% increase in revenue (approximation based on industry standards), the additional revenue generated is ₹1.05 billion (North), dip of ₹0.14 billion (east), and ₹0.28 billion (South), totalling ₹1.19 billion annually, or ₹99.17 million per month.

Next, we compare the costs and benefits of operating 24 warehouses versus 22 warehouses. The incremental warehousing cost for two additional warehouses is ₹780,000. Thus, the total incremental cost is ₹780,000 −₹100,000=₹680,000 per month.

The net benefit is then calculated as

₹99.17million−₹0.68million= ₹98.49million per month.

Thus choosing 24 warehouses is financially superior when compared to using 22 warehouses.

C. Try and answer the questions bothering Arun:

- **Are 22 warehouses more cost-effective than 24?**

  No, while 22 warehouses have lower rental costs, they come with higher freight expenses and a potential drop in service levels, which could lead to lost sales and fulfillment challenges.

- **Should future demand growth be factored in?**

  Yes, demand growth should be considered, as 24 warehouses offer greater flexibility and scalability to accommodate future increases.

- **Can 22 warehouses handle demand elasticity?**

  Unlikely, as rising demand from modern trade and e-commerce may strain the 22-warehouse setup, whereas 24 warehouses provide better responsiveness to demand changes.

  Final Choice: Prioritize 24 warehouses. The service improvements align with Marico's customer-centric strategy, and the cost difference is justified by long-term stability and growth potential

Case 2: Simon's Mall

```
Optimal Store Allocation and Rental Income:
Jewelry: 3 stores
Shoe: 2 stores
Department: 2 stores
Book: 1 stores
Clothing: 3 stores

Total Annual Rental Income: $95250.00
```

Under the base constrains the generated total annual rental income is $95250

```
Optimal solution found (tolerance 1.00e-04)
Best objective 1.012500000000e+05, best bound 1.012500000000e+05, gap 0.0000%
   Total Space  Total Profit  Jewelry Stores  Shoe Stores  Department Stores  \
0        10000       99750.0               3            2                  2
1        11000      101250.0               3            2                  2
2        12000      101250.0               3            2                  2
3        13000      101250.0               3            2                  2

   Book Stores  Clothing Stores
0            1                4
1            2                4
2            2                4
3            2                4
```

Increasing space increases profit when adding 2000 sqft later it plateaus.

	Total Space	Total Profit	Jewelry Stores	Jewelry Profit/Sqft	\
0	9000	95250.0	3	175.0	
1	10000	99750.0	3	175.0	
2	11000	101250.0	3	175.0	
3	12000	101250.0	3	175.0	
4	13000	101250.0	3	175.0	

	Shoe Stores	Shoe Profit/Sqft	Department Stores	Department Profit/Sqft	\
0	2	180.0	2	140.0	
1	2	180.0	2	140.0	
2	2	180.0	2	140.0	
3	2	180.0	2	140.0	
4	2	180.0	2	140.0	

	Book Stores	Book Profit/Sqft	Clothing Stores	Clothing Profit/Sqft	\
0	1	200.0	3	100.0	
1	1	200.0	4	90.0	
2	2	112.5	4	90.0	
3	2	112.5	4	90.0	
4	2	112.5	4	90.0	

	Overall Profit/Sqft
0	141.11
1	133.00
2	125.00
3	125.00
4	125.00

At 11000 sqft the overall profit / sqft decreases but the total profit earned is the highest at 11000 sqft .

Addition of 1000sqft from 9000 sqft yield a increase in profit by $45500. Due to the increase in number of clothing store from 3 to 4. Further increasing of the area which is an infrastructure investment doesn't yield greater profits when considering the investment needed.

As at 11000 sqft the addition of one more book store increases the profit but the book store profit per sqft has dropped by 43%. Which doesn't yield greater profit considering the infrastructure investments we need to make.

Case3: Player signing analysis:

```
Optimal solution found (tolerance 1.00e-04)
Best objective 1.400000000000e+01, best bound 1.400000000000e+01, gap 0.0000%
Optimal solution found:
RS: 1
BS: 0
DE: 1
ST: 1
TS: 1
Total victories: 14.00
```

The above given is the base criteria with the problem defined basic constrains.
sensitivity analysis:

Varying Budget :
from 40 million to 68 million .
The 14-game victory could be attained with a budget of 50 million itself. The $5 million here is the slack.

```
Results for Budget = $50 million:
RS: 1
BS: 0
DE: 1
ST: 1
TS: 1
Total victories: 14.00
```

With a budget of three more million i.e $58 million secures the cubs 16 game victories. Which could increase the playoff chances or a wildcard chance for the cubs. Playoff participation can bring in $10–20 million from ticket sales, merchandise, broadcasting, and concessions. Even a Wild Card appearance can significantly boost the brand and future ticket sales.

```
Solution count 1: 16

Optimal solution found (tolerance 1.00e-04)
Best objective 1.600000000000e+01, best bound 1.600000000000e+01, gap 0.0000%

Results for Budget = $58 million:
RS: 1
BS: 1
DE: 0
ST: 1
TS: 1
Total victories: 16.00
Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (win64 - Windows 10.0 (19045.2))
```

References:

https://www.corporatevision-news.com/how-improving-your-customer-service-levels-drives-revenue/

https://www.bls.gov/news.release/pdf/prod2.pdf